

BIOLOGY 2
TERM 2 EXAM – 2020
MARKING SCHEME

1. What is the role played by the following parts during digestion? (3mks)
 - i) Pancrease – **Secretes hormones insulin and glucagon that regulate blood sugar level.**
 - ii) Liver - **secretes bile stored in the gall bladder. The bile secreted emulsifies fat.**
 - iii) Oesophagus – **Aids in the swallowing of food.**

2. Explain the following;

- i) When transplanting seedling its advisable to remove some of the leaves. (1mk)

To minimize the rate of transpiration.

- ii) Leaves are darker on the upper side than the lower side (1mk)

Presence of numerous chloroplast on the upper side than the lower side.

3. Distinguish between cohesion and adhesion force in the movement of water up the xylem vessels. (2mks)

- **Cohesion is the attraction of water molecules in such a way that they always stick together**
- **Adhesion it is the attraction of water molecules to the walls of the xylem**

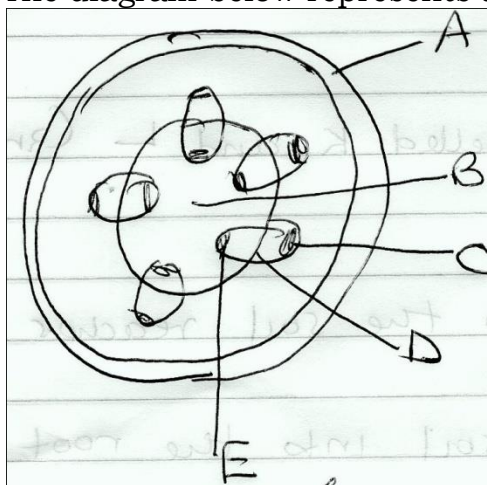
4. Give three adaptation of root hair cell to its function. (3mks)

- i. **Have large vacuole**
- ii. **Long**
- iii. **Thin**
- iv. **numerous**

5. Highlight three roles played by the stem in a plant. (3mks)

- **Support and expose the leaves and flowers to the environment.**
- **Conduct water and mineral ions from roots to rest of plants**
- **Conduct manufactured foods from the [photosynthesis organs to the other parts of the plants.**

6. The diagram below represents a transverse section of a young plant.



- a) From which plant organ was the plant obtained. (1mk)

Dicotyledonous stem

- b) Name the parts labeled A and B. (2mks)

A – Epidermis

**B – Pith**

c) State the functions of part labeled C, D and E. (4mks)

C – Translocates manufactured food /soluble organic products of photosynthesis

D – Divides to give rise to new cells that give rise to secondary phloem and xylem.

E – Transport mineral salts and water from the roots to the leaves.

7. Name two structures where water loss from plants occurs. (2mks)

Stomata

Cuticle

Lenticels

8. Define translocation. (1mk)

It is the transport of soluble products of photosynthesis within the plant.

b) Name the vessel in which translocation occurs in plants. (1mk)

Phloem tissue

9. State two significance of transpiration. (3mks)

- **Mineral ions and water are transported in the plant.**
- **Severs to cool the plants**
- **Responsible for turgor in plants.**

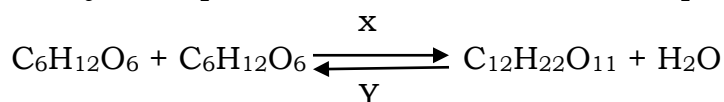
10. Highlight four structural factors that affect transpiration rate. (4mks)

- i. **Cuticle**
- ii. **Leaf size shape**
- iii. **Stomata**
- iv. **Hairy leaves**
- v. **Glossy leaf surface**

11. Name any four forces that are involved in transpiration of water and mineral salts. (4mks)

- **Cohesion and Adhesion force**
- **Capillarity**
- **Root pressure**
- **Transpiration pull**

12. Study the equation below and answer the question that follows.



i) Identify processes x and y. (2mks)

X condensation

Y Hydrolysis

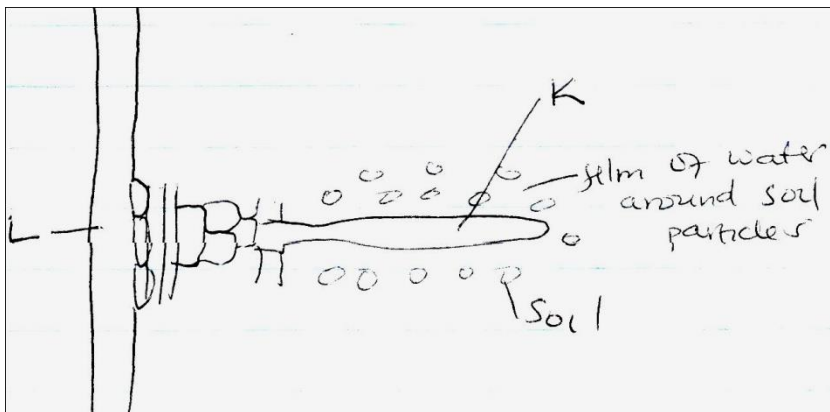
ii) Why are monosaccharide referred to as reducing sugars? (1mk)

Reduce copper ii sulphate into copper I oxide.

iii) Give two other properties of monosaccharide. (2mks)

- **Are sweet**
- **Are crystallisable**
- **From solutions.**

13. The diagram below represents the pathways of water from the soil into the plant.



a) Name the structures labeled K and L. (2mks)

K – Root hair

L – Xylem vessel

b) Explain how water from the soil reaches the structure labeled L. (5mks)

- **Water moves from the soil into the root hair by osmosis**
- **Because concentration of cell sap is higher in the root hair is diluted thus making it less concentrated than neighbouring cell. Water therefore moves into the neighboring cell, it is actively secreted into structure L. (xylem)**

c) Name the process by which mineral salts enter into the plant. (1mk)

- Active transport.

- Diffusion

14. Give the role of the following vascular bundles. (2mks)

Conducts water and mineral salts for the roots to the leaves.

15. Define the following terms as used in Biology Allergens (1mk)

Any substance that causes allergy.